

Foodhub Data Analysis

Python Foundation : Food Hub Data Analysis
PGP-AIML-BA-UTA-Mar25-E

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Executive Summary

Overview

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. Online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company, FoodHub offers access to multiple restaurants through a single smartphone app.

The app allows the restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirms the pick-up in the app and travels to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin of the delivery order from the restaurants.

Executive Summary

Objective

The food aggregator company has stored the data of the different orders made by the registered customers in their online portal. They want to analyze the data to get a fair idea about the demand of different restaurants, which will help them in enhancing their customer experience. Suppose you are hired as a Data Scientist in this company and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company to improve the business.

Executive Summary

- **Conclusion:**

The analysis of Foodhubs data analysis proved key Insight into customer behavior, popular cuisine and restaurants and Operational aspects The below insight and recommendation will help to improve business

- **Insight :**

- American, Japanese, Italian, Chinese and Mexican are the most popular cuisine.
- There are 14 different cuisine ordered by the customer with various
- Cost of the order is right skewed, with 75% of order below \$22.29
- Shake Shack is the most popular restaurant, it accounts for 11.43% of order
- Weekend have 2.5 times more volume compared to weekdays.
- Total cost of the food order was \$31314.82 and 71.51% (\$22392.6) order was on weekend and weekdays it was only 28.49%
- 736 out of 1898 orders don't have ratings and which accounts for 38.78 % of the data
- Food preparation data has multimodal distribution with few peaks around 20, 26 m 30 and 35 min

Executive Summary

Insight

- Food delivery data has multimodal distribution with few peaks around 15 m 29 25 and 278 min.
- There is very low correlation between cost of the order and delivery time or food preparation time and need to probe more and collect data to find impact of these field on revenue and customer experience.
- Average Delivery time on weekend is 22 min vs on weekdays it is 28 days.

Executive Summary

Recommendation

- **Ratings:** A significant gap exists in customer feedback, as most customers do not rate their experience. A focused initiative to actively collect feedback and bridge this gap is needed.
- **Popular Restaurant and Cuisine:** New York have some very popular restaurants and cuisine, and can be considered for targeted marketing and partnership.
- **Weekdays vs Weekend business:** Weekend account for the majority of the business, while weekdays are slow hours. Some strategies focus on weekend promotion to boost weekend sale is needed.
- **Data collection:** We recommend business to collect more data with dates, season, area, distance from restaurants, etc. This will help to establish trends, effectiveness of promotion, optimizing delivery time.
- **Promotion/ Offers:** Increasing margins on food orders less than \$20 will help increase the revenue.
- Some Promotion offer may help to increase the order volume and revenue, but impact of the promotion can't be concluded from the given data.

Business Problem Overview and Solution Approach

Foodhub has collected the food order data from registered customers, Business want to get insight of the data from the data scientist to help improve customer experience, increase revenue and optimize delivery operations.

To analyze the data, following steps will be performed

- Load the data in the Dataframe
- Observe sample rows of the dataset to check if the data is loaded correctly.
- Check their data type of the columns to verify if any formatting, cleaning or typecasting is needed.
- Validate if any data is missing or need any corrective measure like conversion, dropping value or substitute some data.
- Check the statistical summary of the data to get an overview of the numerical fields.
- Perform univalent and data analysis to check the shape, distribution of the data.
- Perform some multivariant data analysis to find correlation among of various fields.

Data Overview

Some sample rows of data

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
1893	1476701	292602	Chipotle Mexican Grill \$1.99 Delivery	Mexican	22.31	Weekend	5	31	17
1894	1477421	397537	The Smile	American	12.18	Weekend	5	31	19
1895	1477819	35309	Blue Ribbon Sushi	Japanese	25.22	Weekday	Not given	31	24
1896	1477513	64151	Jack's Wife Freda	Mediterranean	12.18	Weekday	5	23	31
1897	1478056	120353	Blue Ribbon Sushi	Japanese	19.45	Weekend	Not given	28	24

Data Overview

Statistical summary of the data

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
order_id	1898.0	NaN	NaN	NaN	1477495.5	548.049724	1476547.0	1477021.25	1477495.5	1477969.75	1478444.0
customer_id	1898.0	NaN	NaN	NaN	171168.478398	113698.139743	1311.0	77787.75	128600.0	270525.0	405334.0
restaurant_name	1898	178	Shake Shack	219	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cuisine_type	1898	14	American	584	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cost_of_the_order	1898.0	NaN	NaN	NaN	16.498851	7.483812	4.47	12.08	14.14	22.2975	35.41
day_of_the_week	1898	2	Weekend	1351	NaN	NaN	NaN	NaN	NaN	NaN	NaN
rating	1898	4	Not given	736	NaN	NaN	NaN	NaN	NaN	NaN	NaN
food_preparation_time	1898.0	NaN	NaN	NaN	27.37197	4.632481	20.0	23.0	27.0	31.0	35.0
delivery_time	1898.0	NaN	NaN	NaN	24.161749	4.972637	15.0	20.0	25.0	28.0	33.0

Data Overview

Ans 1: Food Order Data sheet have 1898 rows and 9 columns.

Ans 2: Data sheets have 1 Float, 4 integers and 4 Object data types columns

The Following Columns are present in the Food Order Data sheet,

Columns Name	Data Type	Remarks
order_id	int64	Unique ID of order
customer_id	int64	Unique customer ID will be useful to find not frequent customer
restaurant_name	object	Name of the restaurant will be useful to see popular restaurants, ratings of the restaurants
cuisine_type	object	Cuisine type of useful to see popular cuisine, ratings, cost
cost_of_the_order	float64	cost of the order is the main driver for the business revenue
day_of_the_week	object	Day of the week to find the trend, behavior etc
rating	object	Rating indicate customer experience
food_preparation_time	int64	Time it takes for the restaurant to prepared food on receipt of the order and handover to delivery
delivery_time	int64	Time it takes to deliver food to customer after received from restaurant

Data Overview

Ans 3: There are no missing value in the data sheet, No action is needed

Ans 4: Food Preparation takes minimum **20** minutes average it takes **27.37** minutes and maximum **35** minutes

Ans 5: **736** orders are not rated.

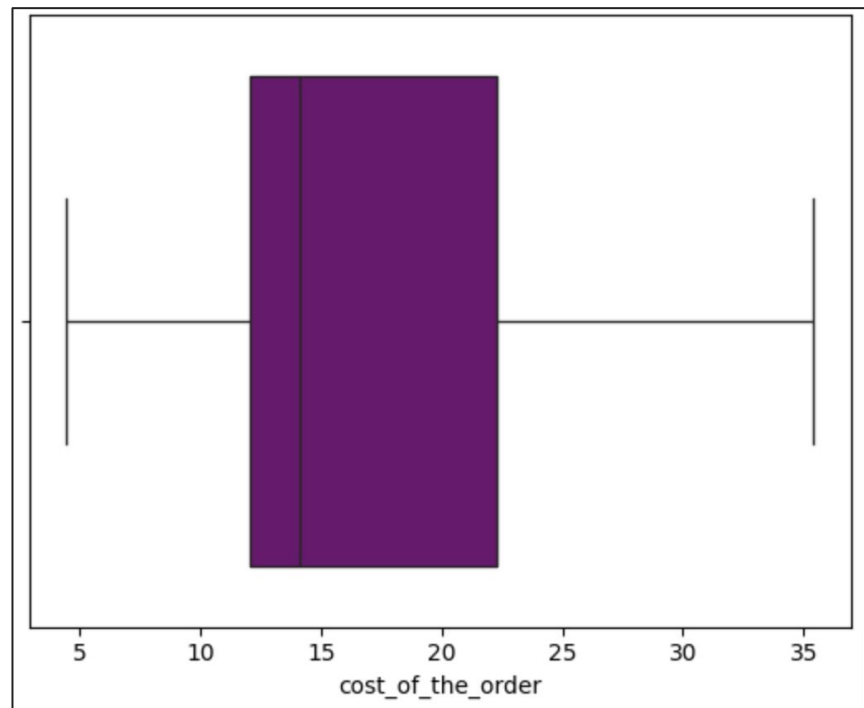
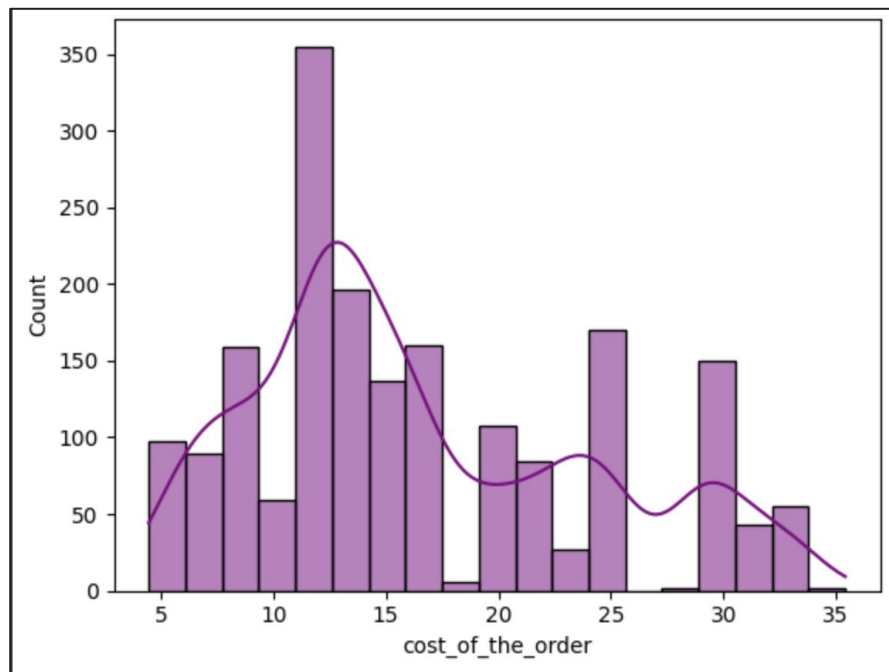
Univariate Analysis

Customer_id and Order_id are numerical column and unique identifier of entity. These columns will not add value in data analysis, so will skip them from Univariate Analysis.

Columns Name	Data Type	Remarks
order_id	int64	Unique Identifier
customer_id	int64	Unique Identifier
restaurant_name	object	Large cardinality and categorical column
cuisine_type	object	categorical column
cost_of_the_order	float64	Numeric
day_of_the_week	object	categorical column
rating	object	categorical column
food_preparation_time	int64	Numeric
delivery_time	int64	Numeric

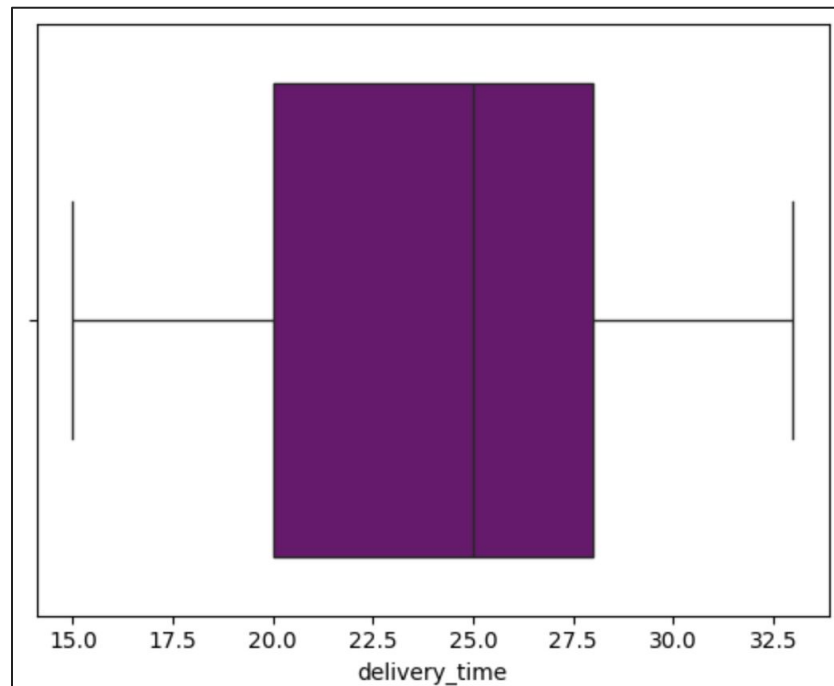
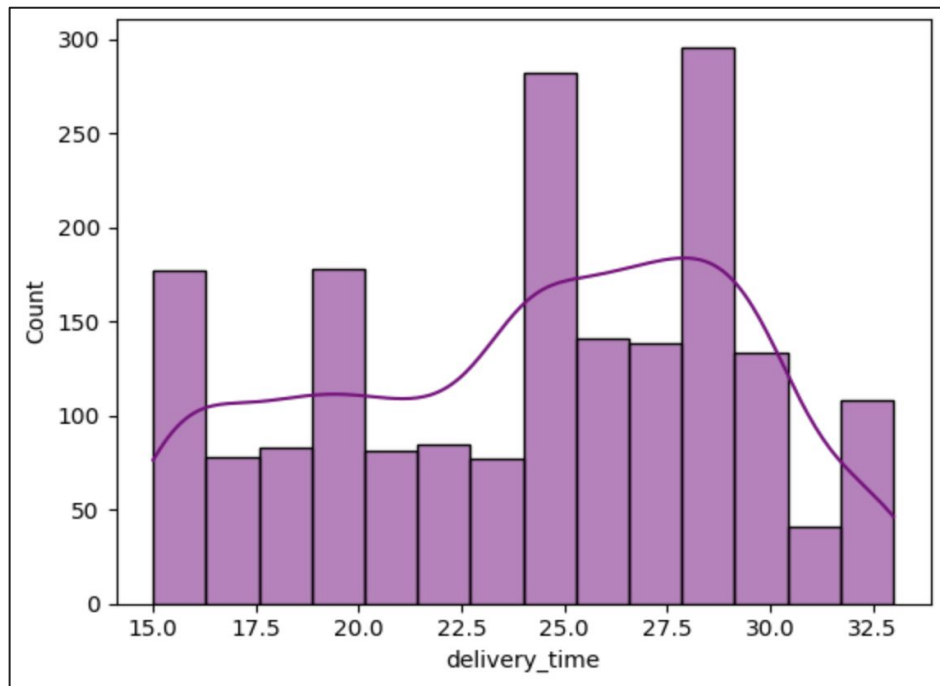
Univariate Analysis

- **cost_of_the_order:** Data is right skewed, with mean at \$16.50 with minimum order of \$ 4.47 and max at \$35.41. 75% order cost below \$22.50



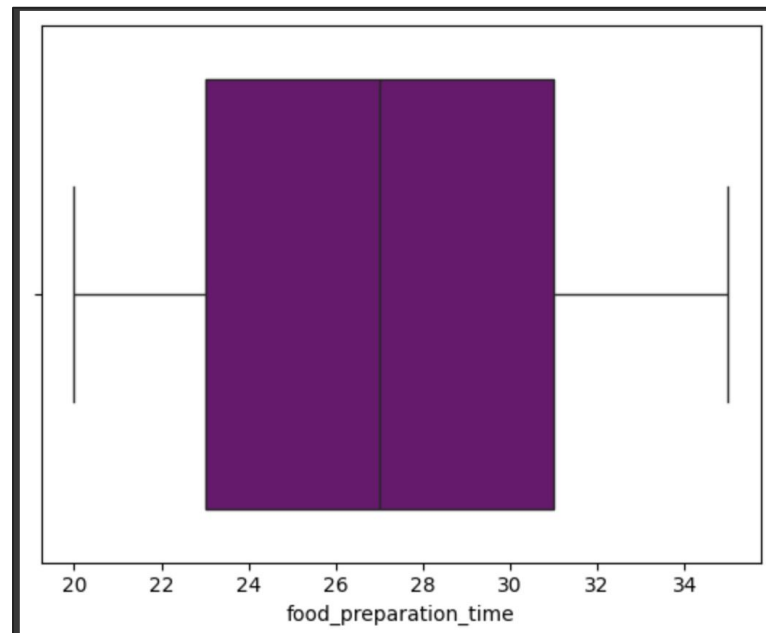
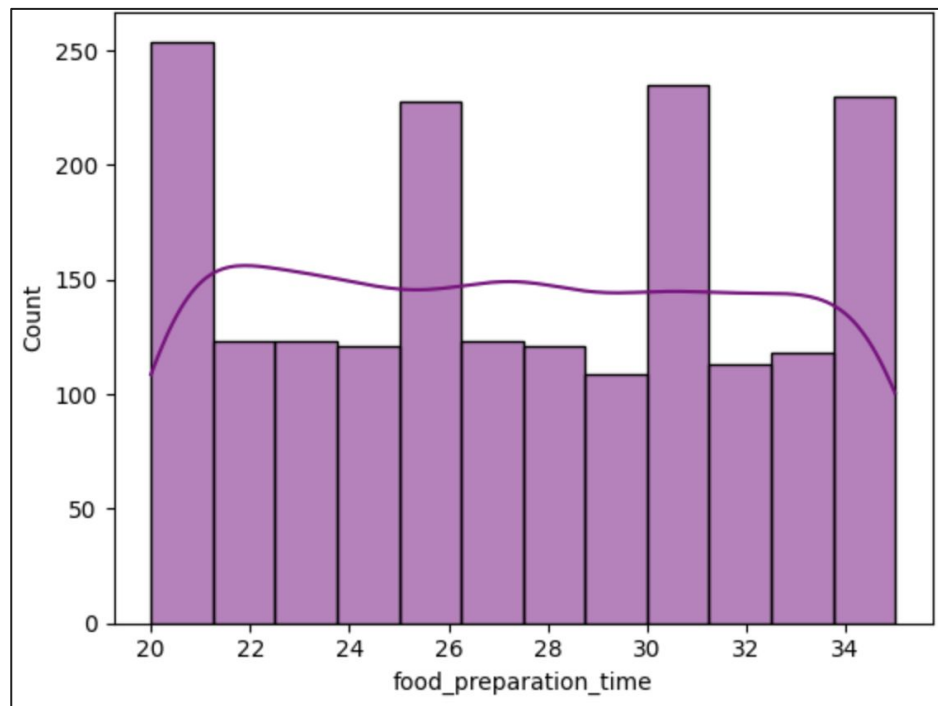
Univariate Analysis

- delivery_time**: is numerical column with mean 24.16 min and 75% order takes less than 28 min
Data is multimodal with multiple peak as 15, 20, 25 and 30 min, data is left skewed it



Univariate Analysis

- **Food_preparation_time**: is numerical column, and it is right skewed mean food preparation time is 27.37 min



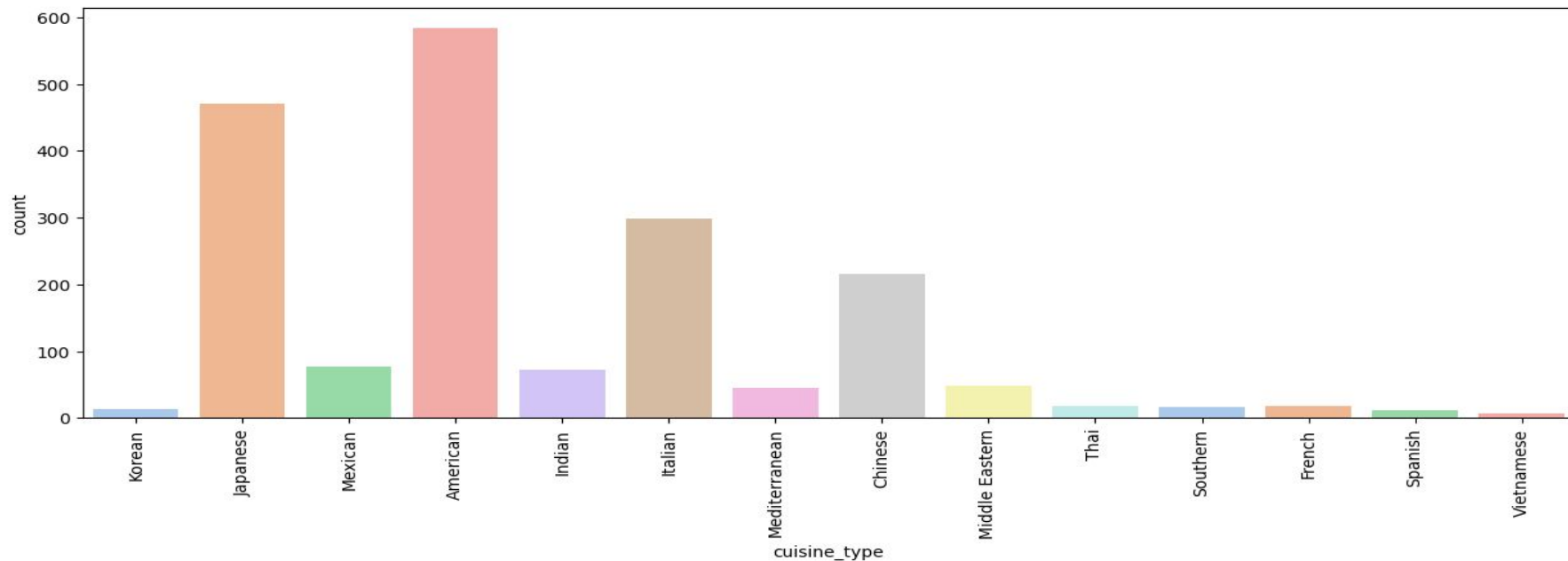
Univariate Analysis

restaurant_name: Food Order dataset have 178 Restaurants and Below are 5 top restaurants by order counts.

Restaurant Name	Number of orders
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

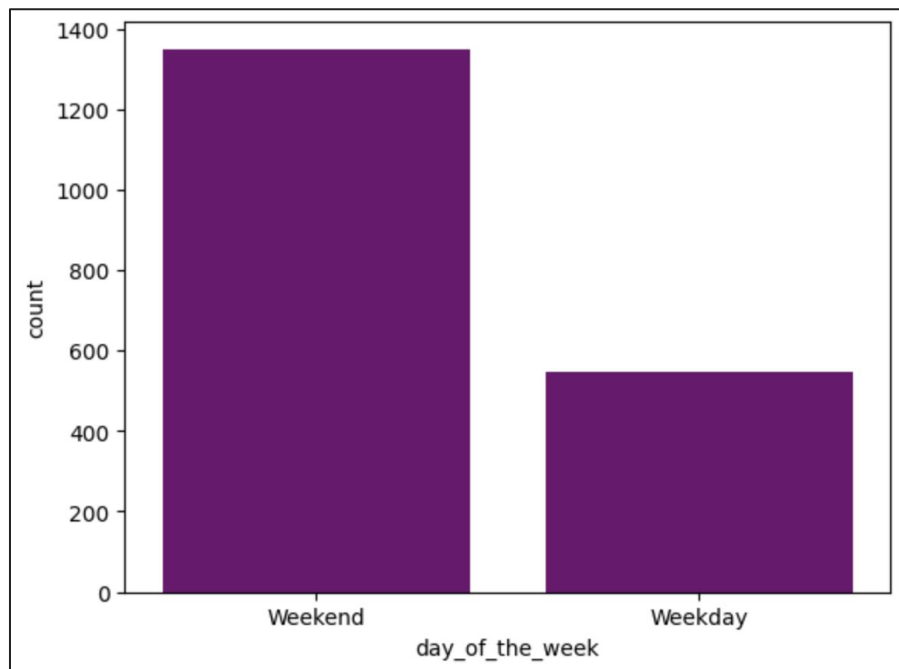
Univariate Analysis

- **cuisine_type:** It is a categorical variable and there are total 14 unique different cuisine_type in the dataset. American, Japanese, Italian, Chinese, Mexican are the most popular cuisine



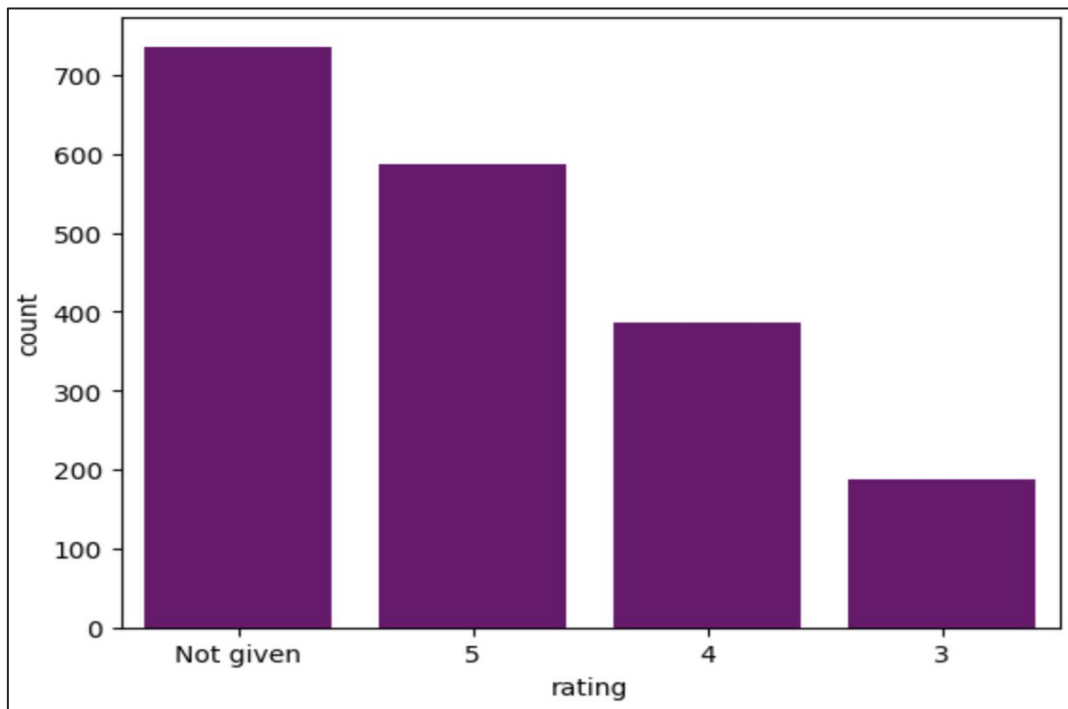
Univariate Analysis

- `day_of_the_week`: is a categorical column, with `weekend` and `weekday` are two values. Weekend have 1351/1898 order that is 71% of orders are placed on the weekend. On Weekdays 547/1898 order placed is 29% order



Univariate Analysis

- rating: is categorical variable with **736** order have no ratings, Not rated account for 38% of the data and need some business effort to address this gap. None of the order have 1 or 2 ratings and need some more investigation



Univariate Analysis

Ans 7: Below are the top 5 restaurants by most numbers orders received

Restaurants	Number of orders Received
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

Univariate Analysis

Ans 8: American Cuisine is the most popular cuisine on the weekends.

Ans 9: 29.24 % of orders are above 20 dollars.

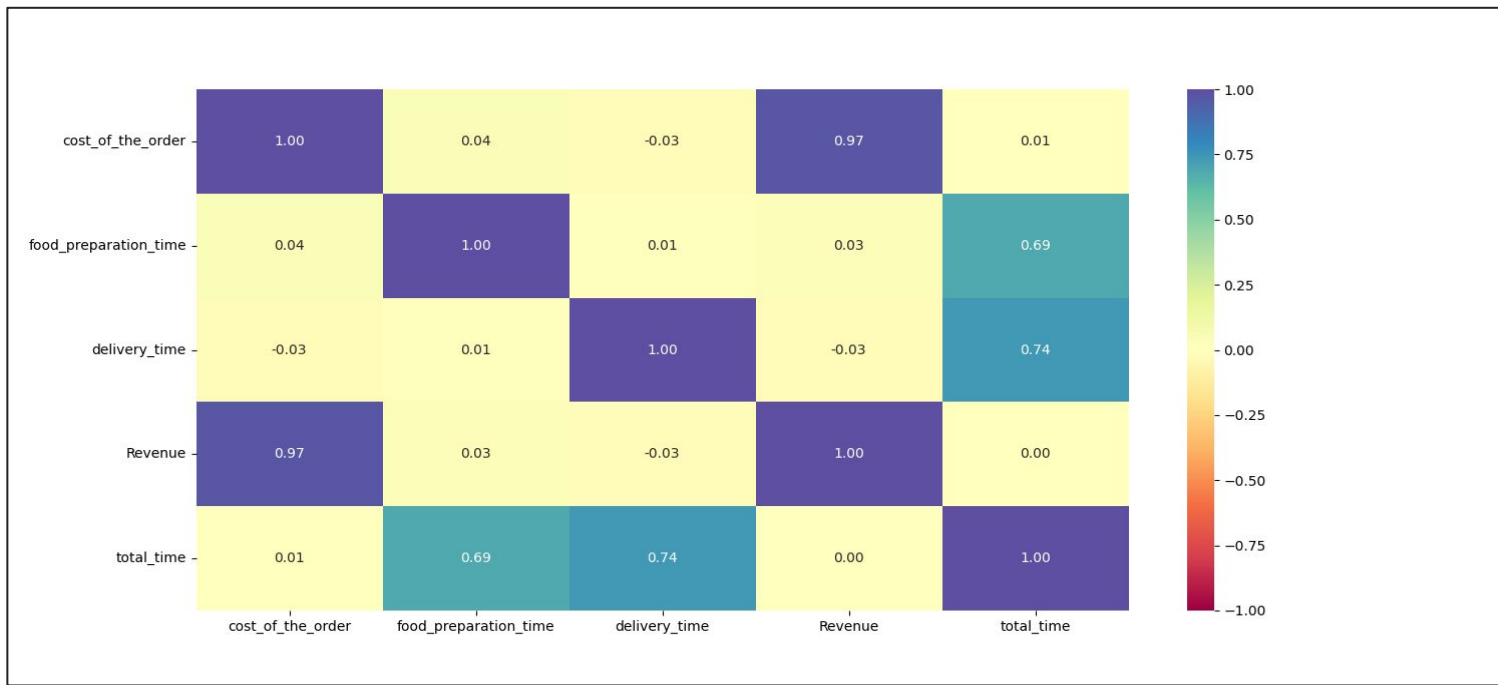
Ans 10: The mean delivery time for this dataset is 24.16 minutes.

Ans 11: Below are the 5 most frequent customer IDs and eligible for 20% discount voucher.

#	most frequent Customer Ids are
1	52832
2	47440
3	83287
4	250494
5	259341

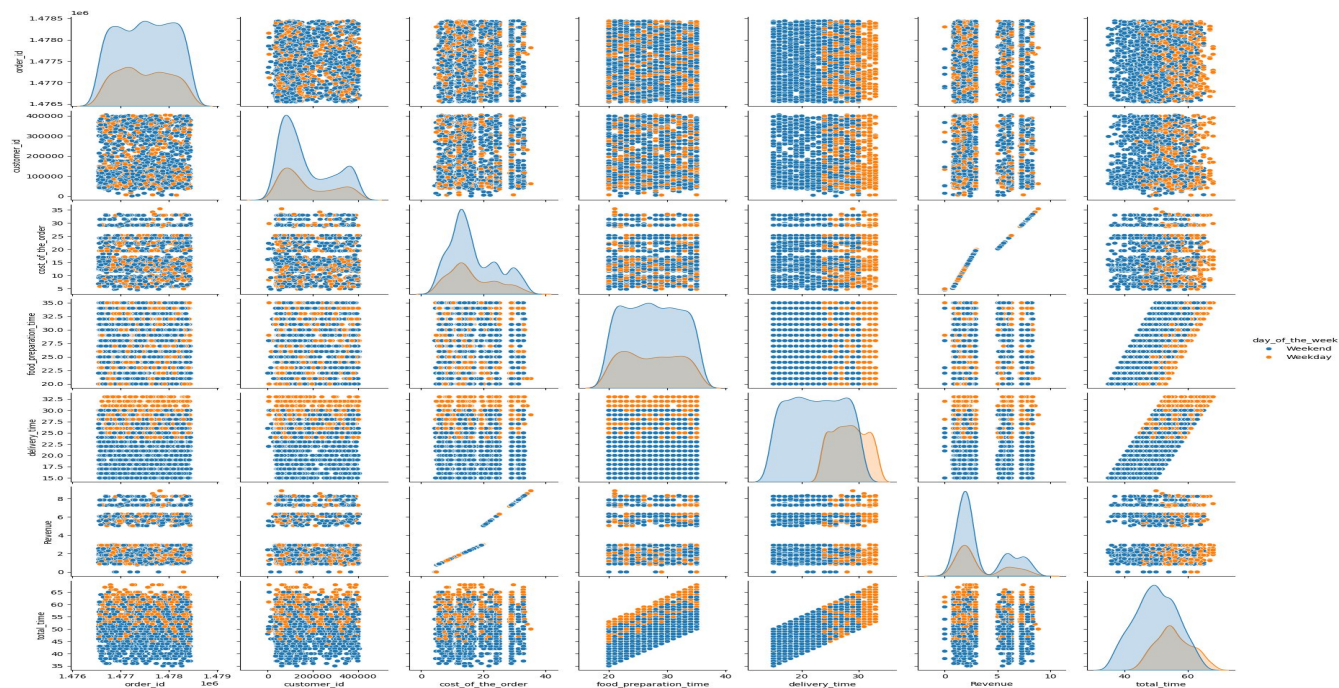
Multivariate Analysis

- Will analyze relation between two or more fields. Using heatmap, pair plot etc.



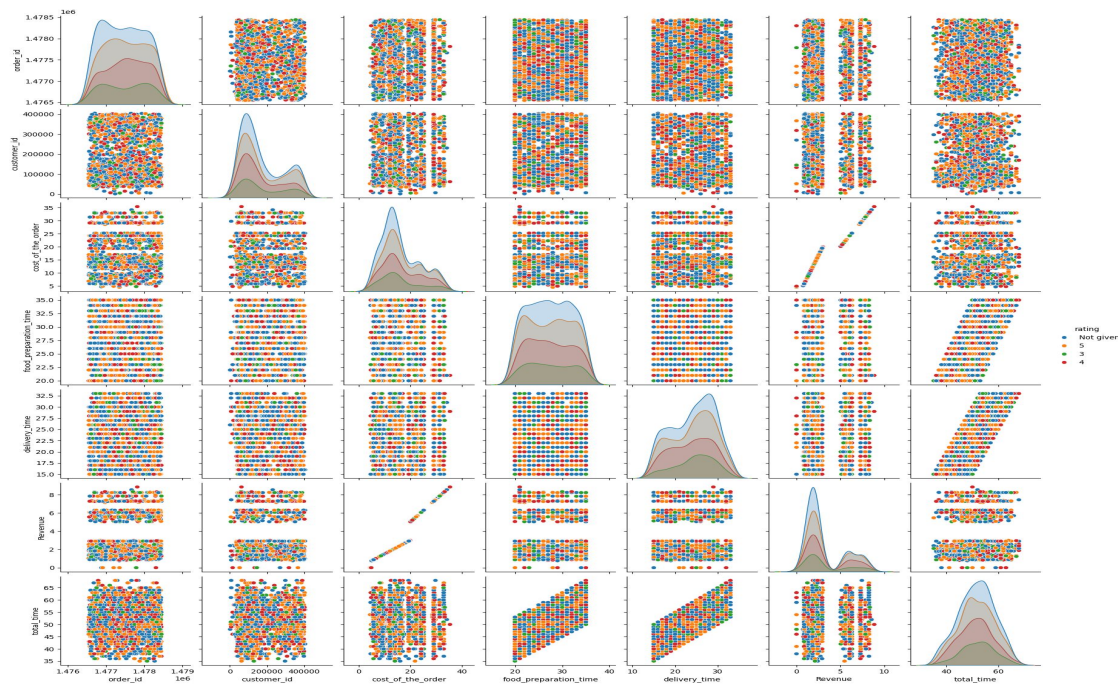
Multivariate Analysis

pair plot with Day of the week grouping



Multivariate Analysis

pair plot with group by rating



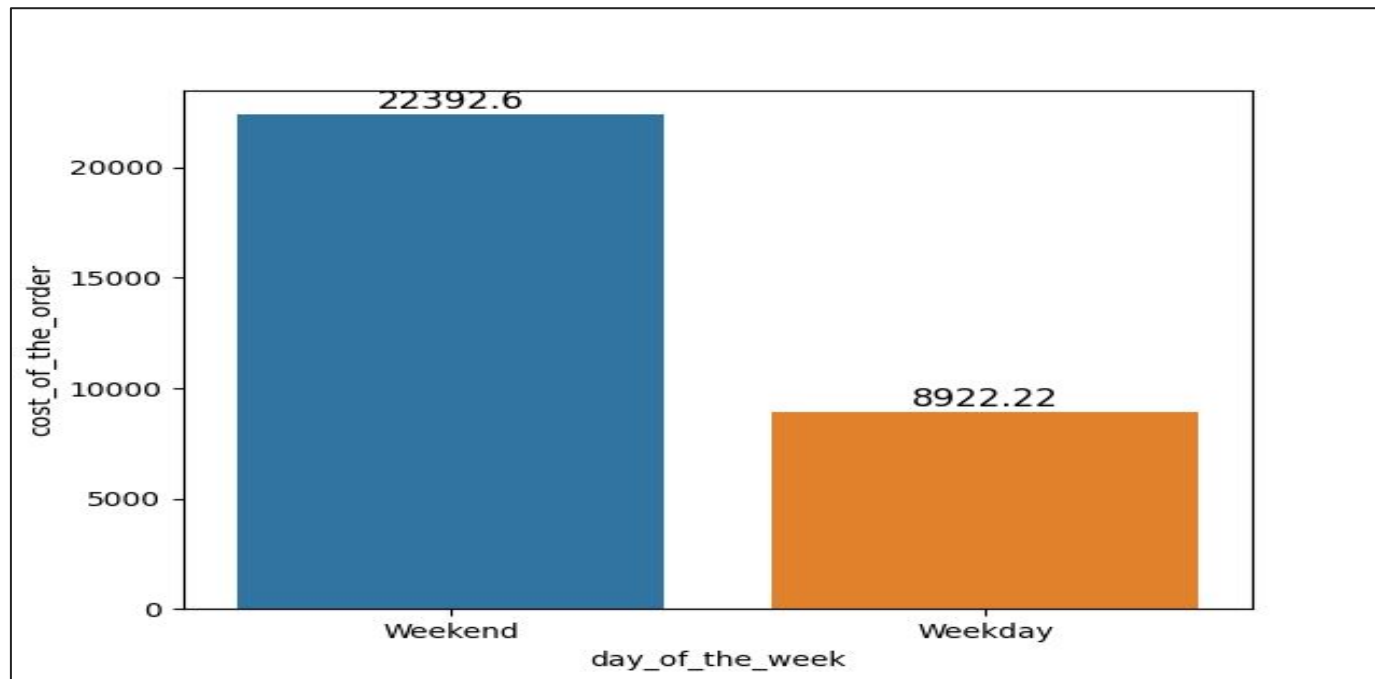
Multivariate Analysis

Both pair plots and heat map have same conclusion that

1. total_time is sum of food_preparation and delivery_time and show high correlation
2. Revenue is directly related to the cost of the order, so it also shows high correlation
3. food_preparation and cost_of the order seems to have very weak correlation
4. delivery_time and cost_of the order seems to have very weak correlation

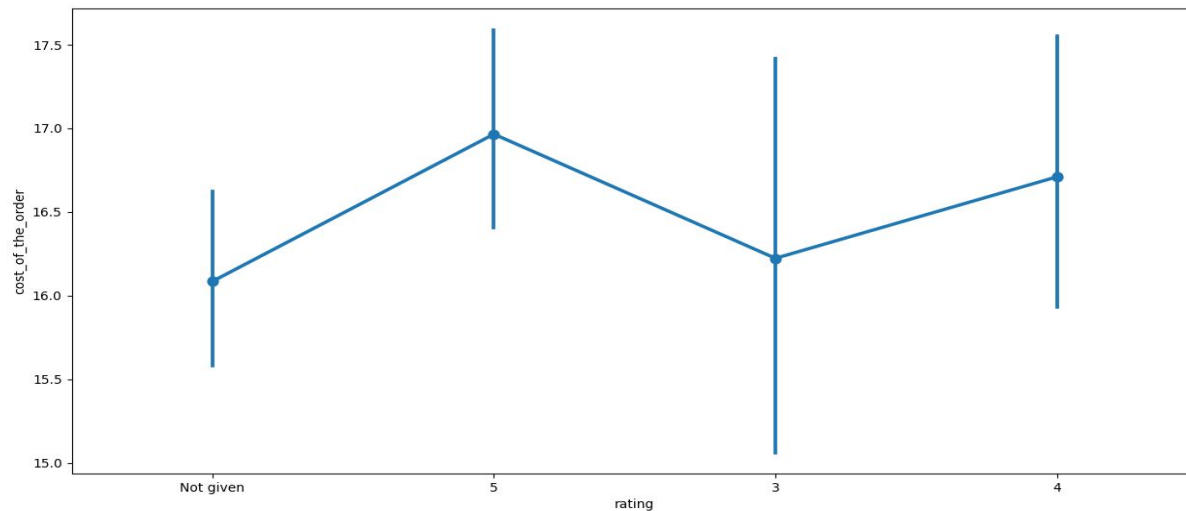
Multivariate Analysis

cost_of_the_order and day_of_the week shows that Weekdays and weekend have similar pattern, but volume of the order is very less compare to weekdays



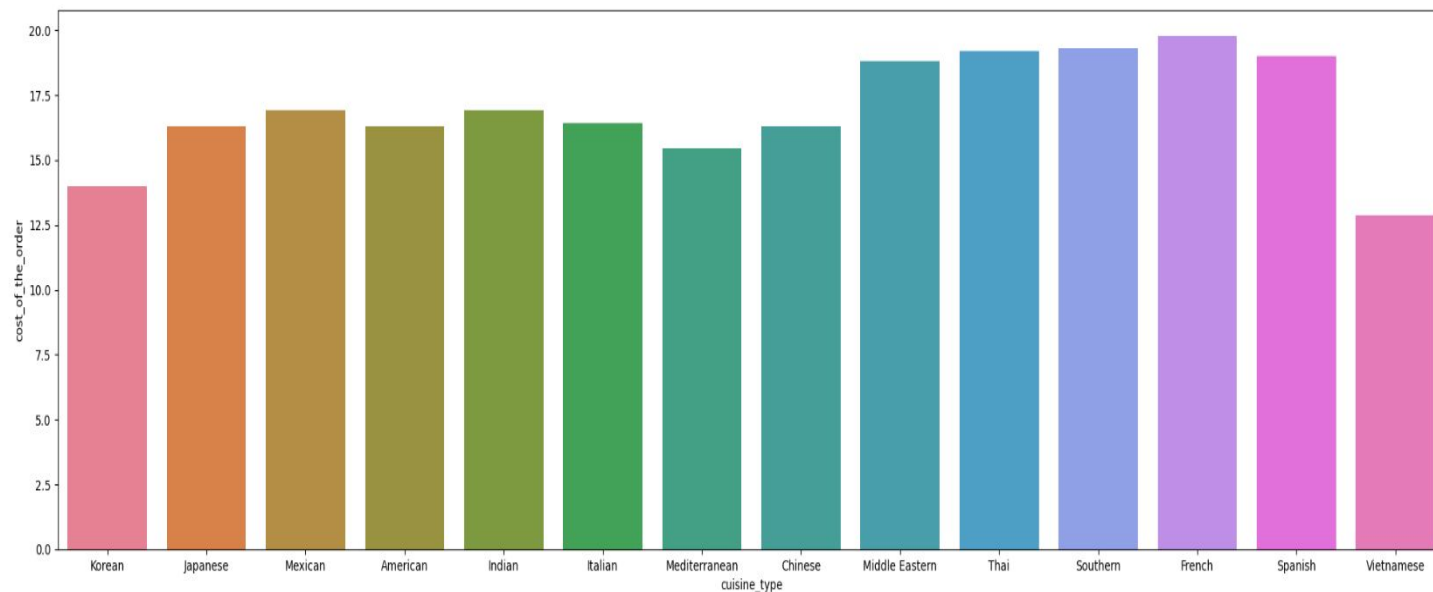
Multivariate Analysis

cost_of_the_order and ratings don't show any relationship and there is no conclusion can be drawn from it



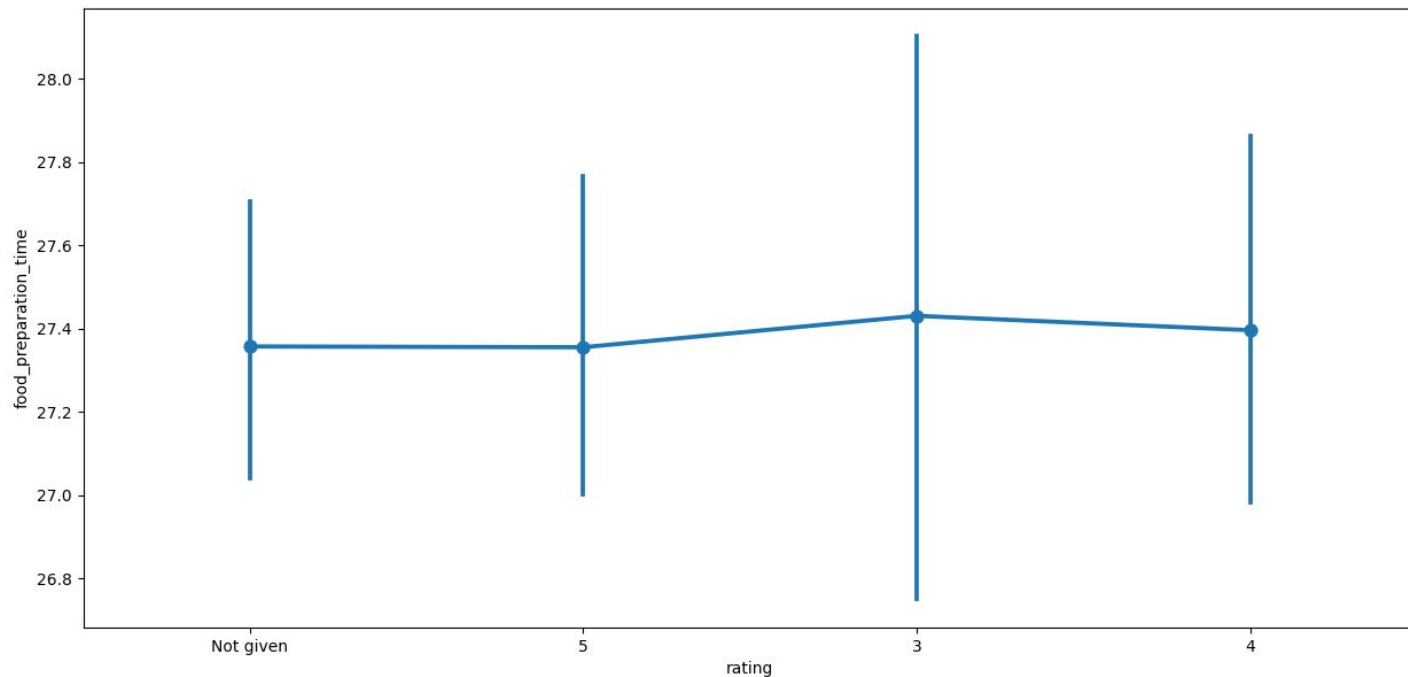
Multivariate Analysis

cost_of_the_order and cuisine_type also don't have any relationship. Some cuisine have higher mean value, but it could be due to their low volume and range of cost_of_the_order



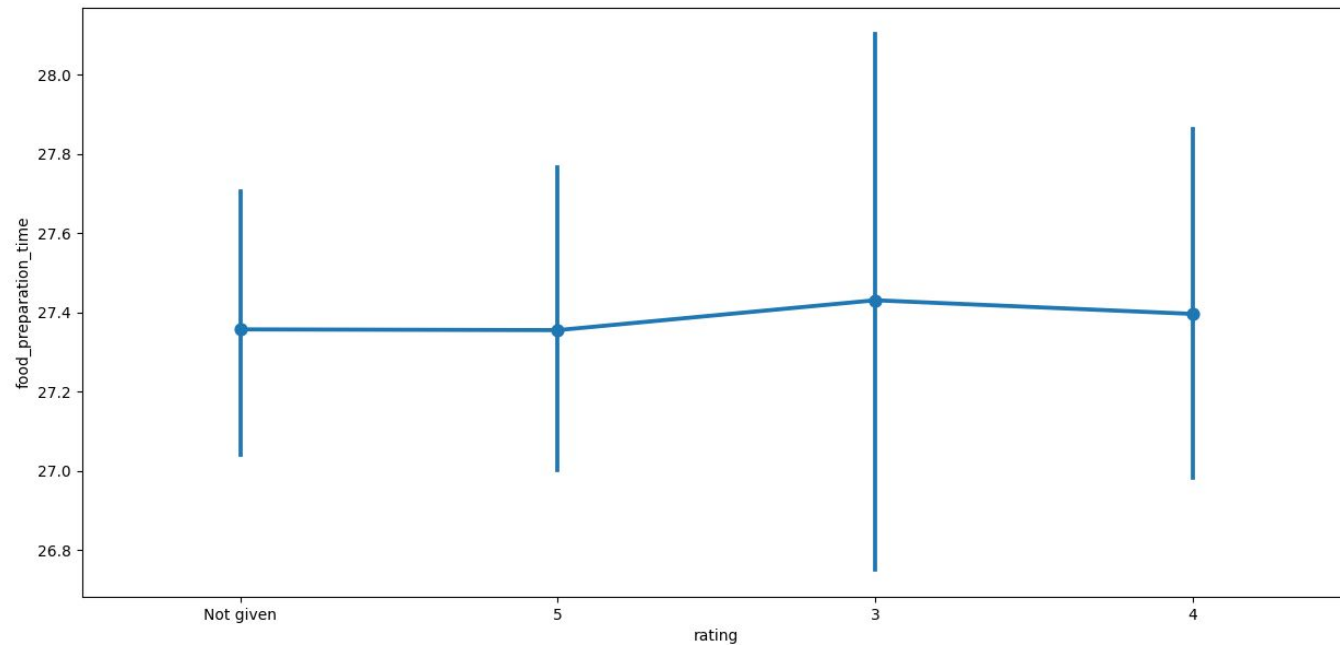
Multivariate Analysis

rating and food_preparation_time also don't show any relationship



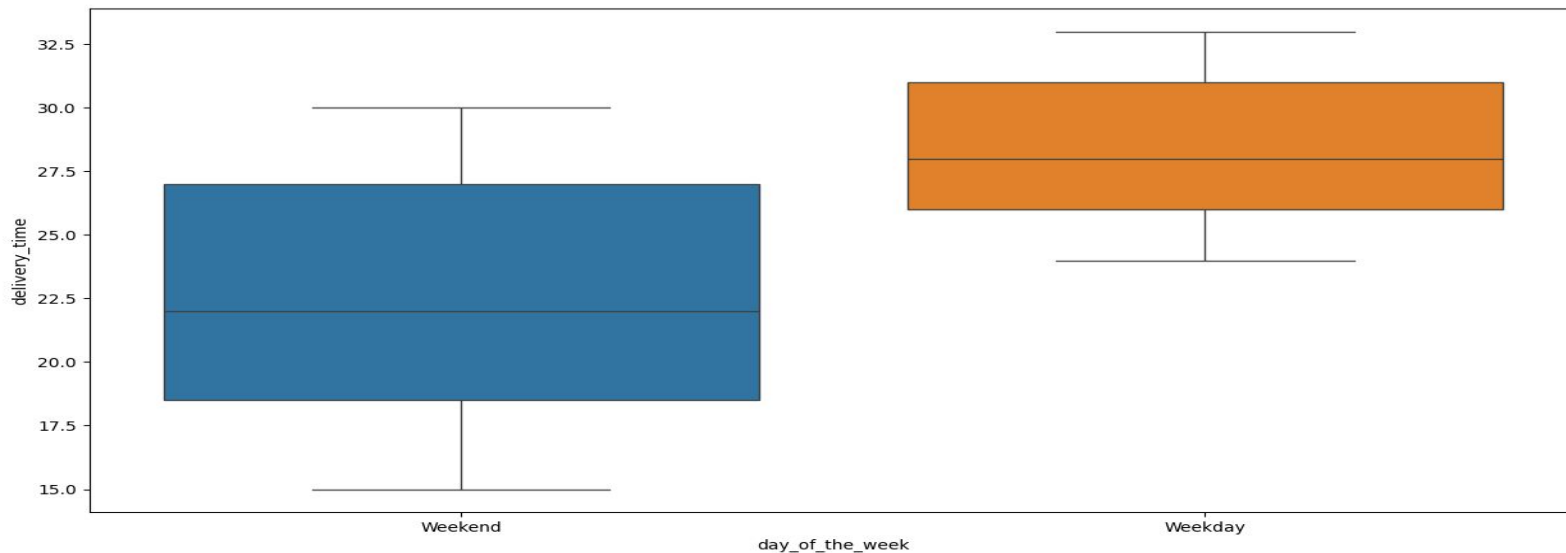
Multivariate Analysis

delivery_time has little or no impact on rating



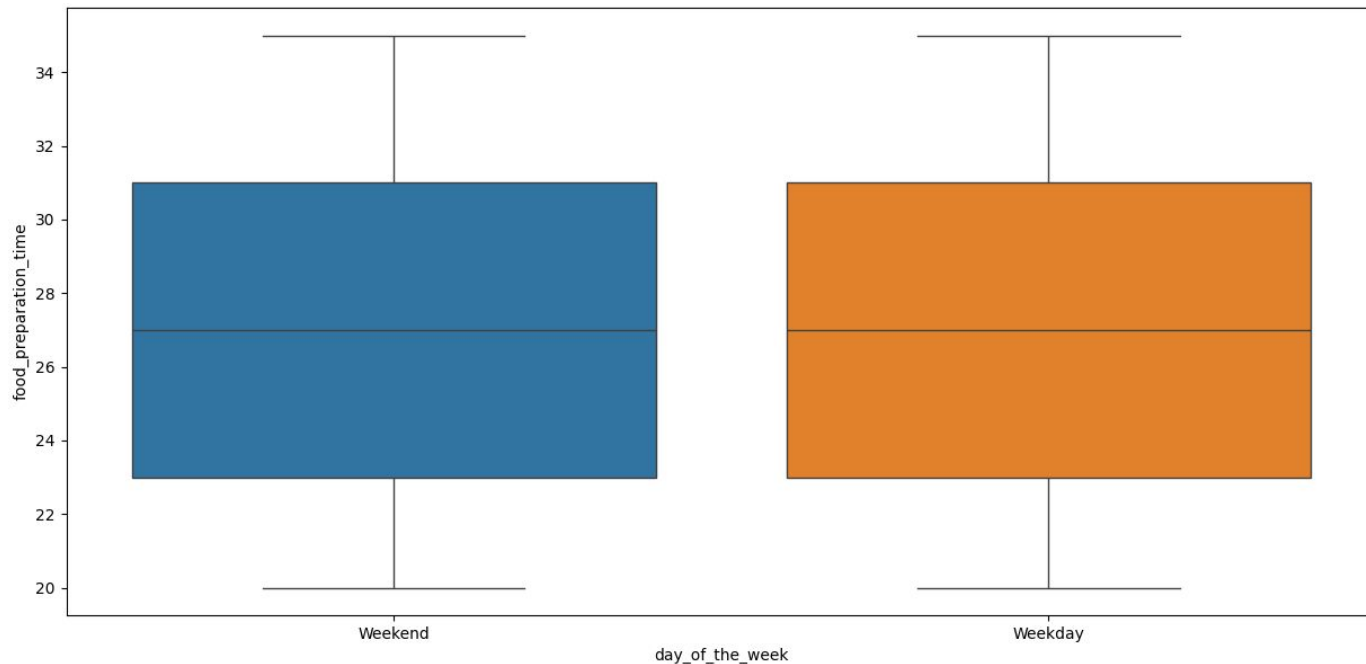
Multivariate Analysis

delivery_time and day_of the week : On weekend it takes less amount of time compare to weekdays even though volume of the order is less on weekdays. And should be investigated further



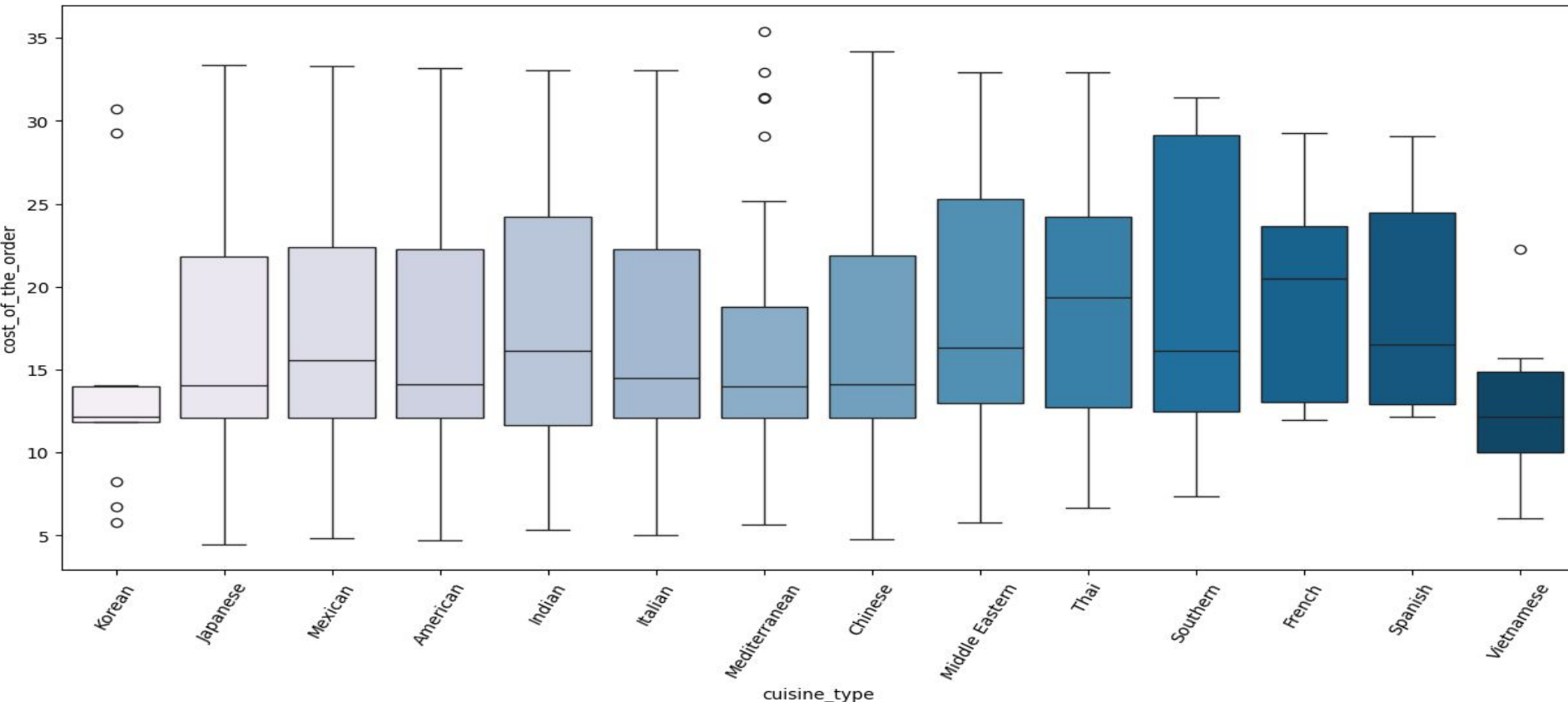
Multivariate Analysis

food_preparation_time and day_of the week : day of the week don't impact food_preparation_time



Multivariate Analysis

Cuisine vs cost_of_the_order: Cuisine have little impact on cost of the order, few cuisines have some lower mean compare to others.



Multivariate Analysis

Ans 13: Below are the restaurants that have more than 50 review with average ratings more than 4. A promotional offer in the advertisement for the restaurants will be good for business as this is highly praised by the customer and very popular too.

Restaurant names	Rating (descending Order)
The Meatball Shop	4.51
Blue Ribbon Fried Chicken	4.33
Shake Shack	4.28
Blue Ribbon Sushi	4.22

Multivariate Analysis

Ans 14: The net revenue is around 6,166.3 dollars.

Ans 15: 10.54% of the orders cost more than 60 minuter. Some more investigation needed to find the reason for high total time it took for these order some additional data points like distance. Or complexity of the order will be useful.

Ans 16:

The mean delivery time on weekdays is around 28 minutes.

The mean delivery time on Weekends is around 22 minutes.

APPENDIX



Happy Learning !

